

ETHERNET STANDARD AND SPECIFICATION

1. RJ-45 Pin Layout and Color Codes

Pin	T568A Color	T568B Color
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

RJ-45 Connector (Front View)

 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

2. Comparison of Ethernet Standards

Standard	Speed	Cable Type	Maximum Length	Connector
10BASE-T	10 Mbps	Cat3/Cat5	100 m	RJ-45
100BASE-TX	100 Mbps	Cat5	100 m	RJ-45
1000BASE-T	1000 Mbps (1 Gbps)	Cat5e/Cat6	100 m	RJ-45

3. Connecting Two Laptops Directly

Cable Used: Crossover Cable

Reason: A crossover cable swaps the transmit (TX) and receive (RX) wire pairs, allowing two similar devices (laptop-to-laptop) to communicate directly without a switch or hub.

4. Real-World Devices Using Ethernet

Device	Likely Ethernet Standard
Wi-Fi Router	1000BASE-T
Desktop Computer (College Lab)	1000BASE-T
Network Printer	100BASE-TX

Explanation

- Wi-Fi Router: Uses Gigabit Ethernet for high-speed internet.
- Desktop Computer: Usually supports Gigabit Ethernet for fast network access.
- Network Printer: Often uses 100 Mbps Ethernet, which is sufficient for printing tasks.